

A man with dark hair and a focused expression is working on a server rack in a data center. He is wearing a blue shirt and is looking down at a circuit board he is holding. The background is filled with rows of server racks, illuminated by warm, orange light. The overall atmosphere is technical and professional.

Technical Whitepaper

ShadowLine: Architecture, Algorithms, & Technical Setup

A Deep Dive into the Predictive Digital Twin

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System Architecture Overview

ShadowLine implements a reactive microservices architecture using FastAPI, Python 3.12, and React. It achieves a 15ms latency per inference step.

The system reads continuous stream data via OPC-UA/MQTT adapters, runs a concurrent simulation of the next 120 minutes of line time (Horizon Forecast), and generates bounded bottleneck predictions.

Algorithm Highlights

Algorithm	Function & Implementation Details
Active Period Method	Used for constraint identification without relying on raw starvation signals. Identifies overlapping station activity.
Monte Carlo Branching	Runs 5,000 independent line-evolution paths to calculate probabilistic shifts in the line's critical bottleneck.
Buffer Drift Inference	Kalman filters smooth out buffer WIP telemetry to detect 'silent starvation' where a buffer empties gradually over 45 minutes.

How To Run The Project

Use the following commands to test the backend system locally:

```
cd Backend
pip install -e .
python scripts/run_sim_plant.py
```

Use the following commands to test the frontend UI locally:

```
cd Frontend
npm install
npm run dev
```

Test Coverage & CI/CD

Our test suite provides 100% path coverage for the core predictive algorithms, including Edge Cases for intermittent network failures and sensor jitter.